



—Part 3

MyISAM is MySQL's default storage engine, and is probably most commonly used by new adopters of MySQL. The objective of this article is to discuss how the MyISAM storage engine works, to enable developers to write better SQL queries.

Before we jump into the complex detailing of MyISAM, it's time to learn some of its important characteristics:

- MyISAM stores each table in two files—one containing data, and the other, indexes. The extensions of these two files' names are *.MYD* and *.MYI* respectively. Using the *DATA DIRECTORY* and *INDEX DIRECTORY* options to the *CREATE TABLE* query, you can put the above mentioned two files in different directories. These two directories, put on different disks, can offer significant performance improvement. This technique is often used in case of very heavily loaded databases, and is supported by many databases.
- The data stored in these files is platform-independent—or, more technically put, is not dependent on the endianness of a machine. In other words, you can simply copy (literally) data and index files from an Intel x86/x64 machine to PowerPC or Sun SPARC.
- MyISAM doesn't support transactions. Only table locks are available. Row-level locks, which are a must for MVCC capabilities (transactions, to put it in simple words) are not available in MyISAM.
- All the readers (with *SELECT* family of queries) have to obtain read locks. Multiple users can acquire read

locks at the same time; hence, the common term *shared locks* is frequently used.

- All the writers have to obtain exclusive locks. In simple words, two users cannot perform writes (*INSERT*, *UPDATE*, *DELETE* family of operations) at the same time. This is very important to remember. A write operation must finish very quickly. Otherwise, in case of a database with a heavy load, there will be a long queue of connections waiting for exclusive (write) lock. In general, write operations are not time-consuming; however, the indexing strategy plays a role here. More on this later.
- One great feature offered by MyISAM is called concurrent inserts. These allow the insertion of new records while *SELECT*-family queries are running. However, as insertion requires an exclusive lock, only one user can perform concurrent inserts.
- MyISAM supports full-text index (Full-Text Search, which we will discuss later in this series).
- MyISAM supports compressed tables. When a table is compressed using the *mysampack* utility, individual rows are compressed and not the entire table as a single entity. Please remember that compressed tables are read-only. When a *SELECT* query is issued on a compressed table, rows are decompressed based on the requirement. A